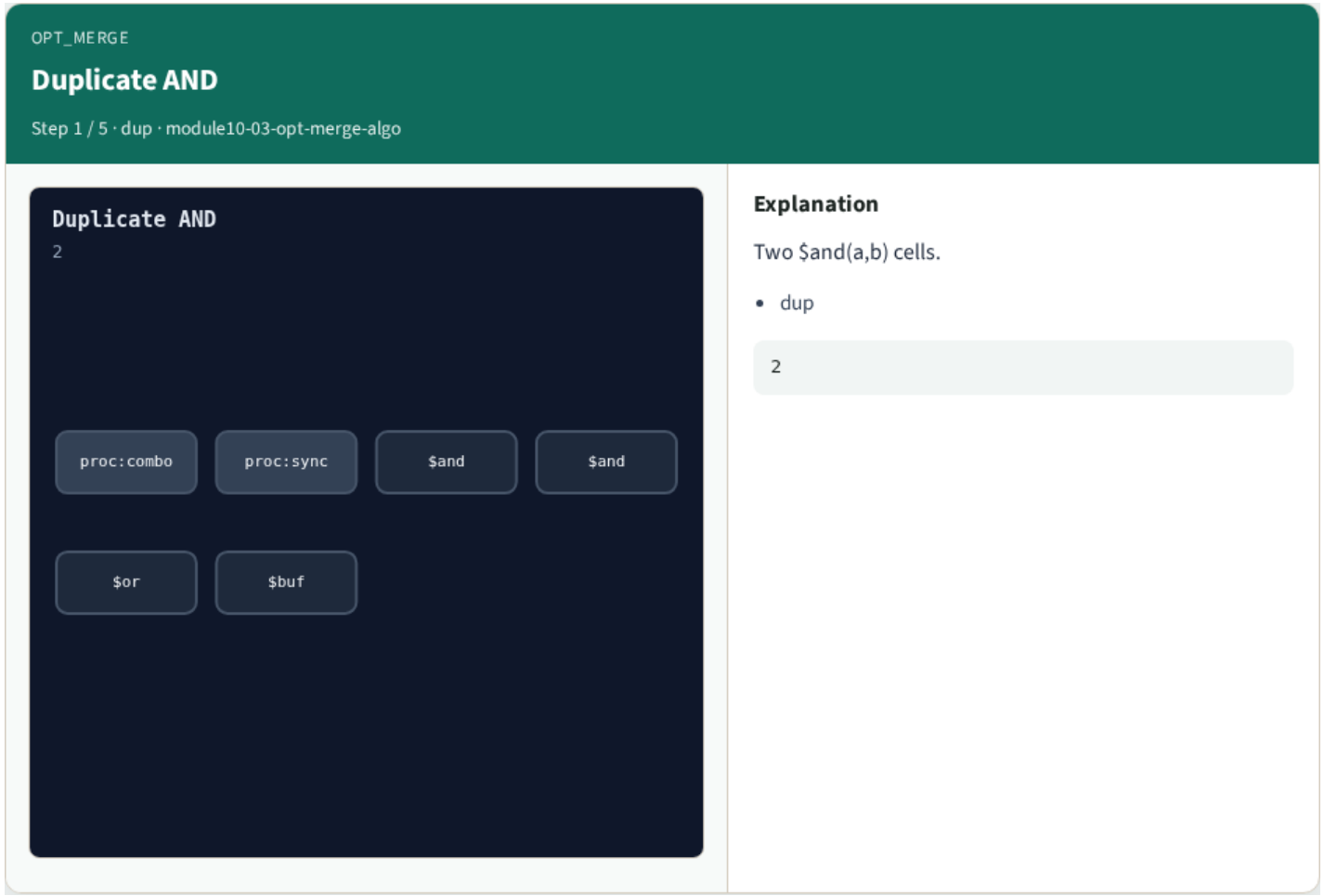
Why opt merge

Identical gates sharing inputs are the same node; merge retargets fans

The idea

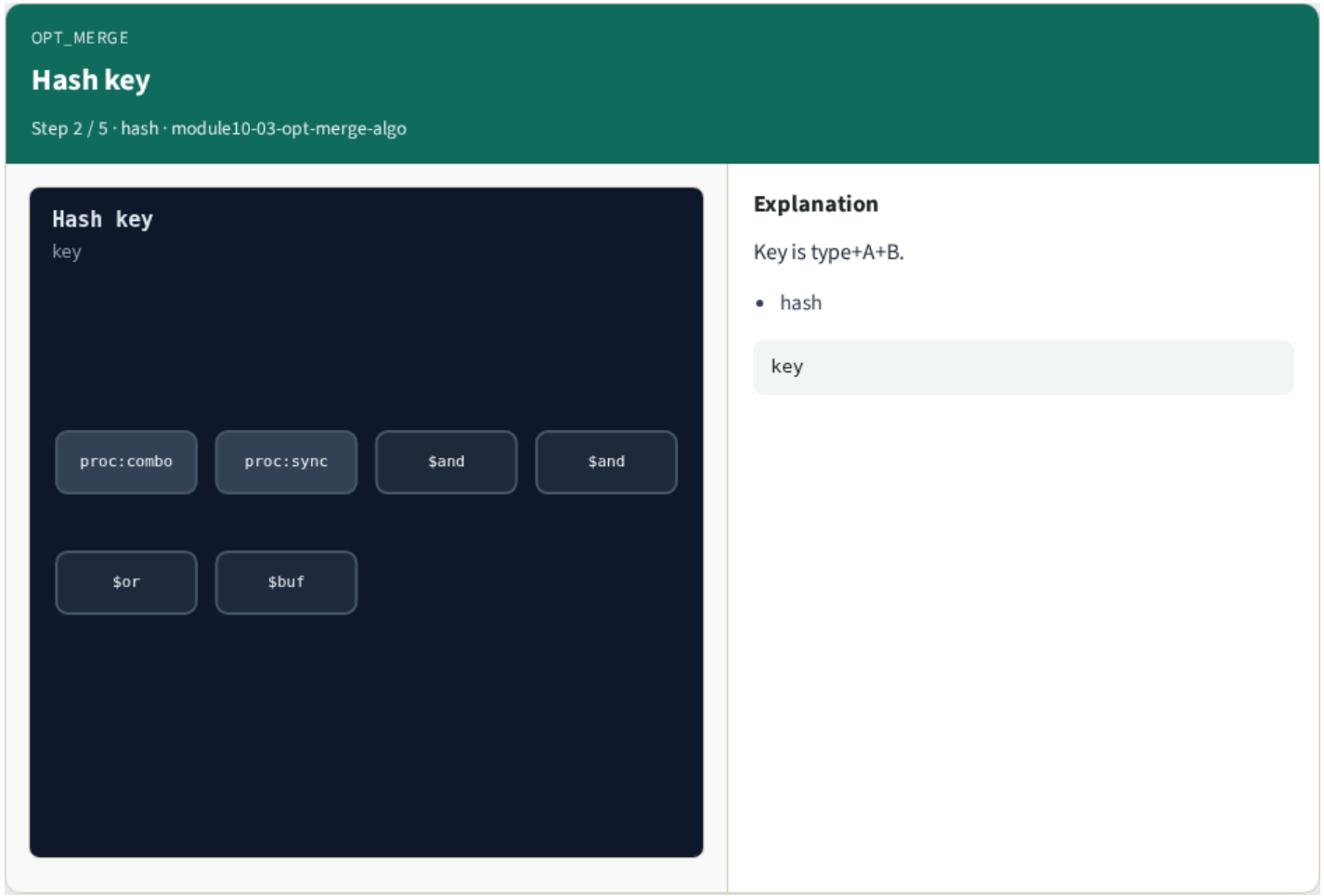
- GOLDEN: after merge and_count=1 (one duplicate AND removed)

Duplicate AND



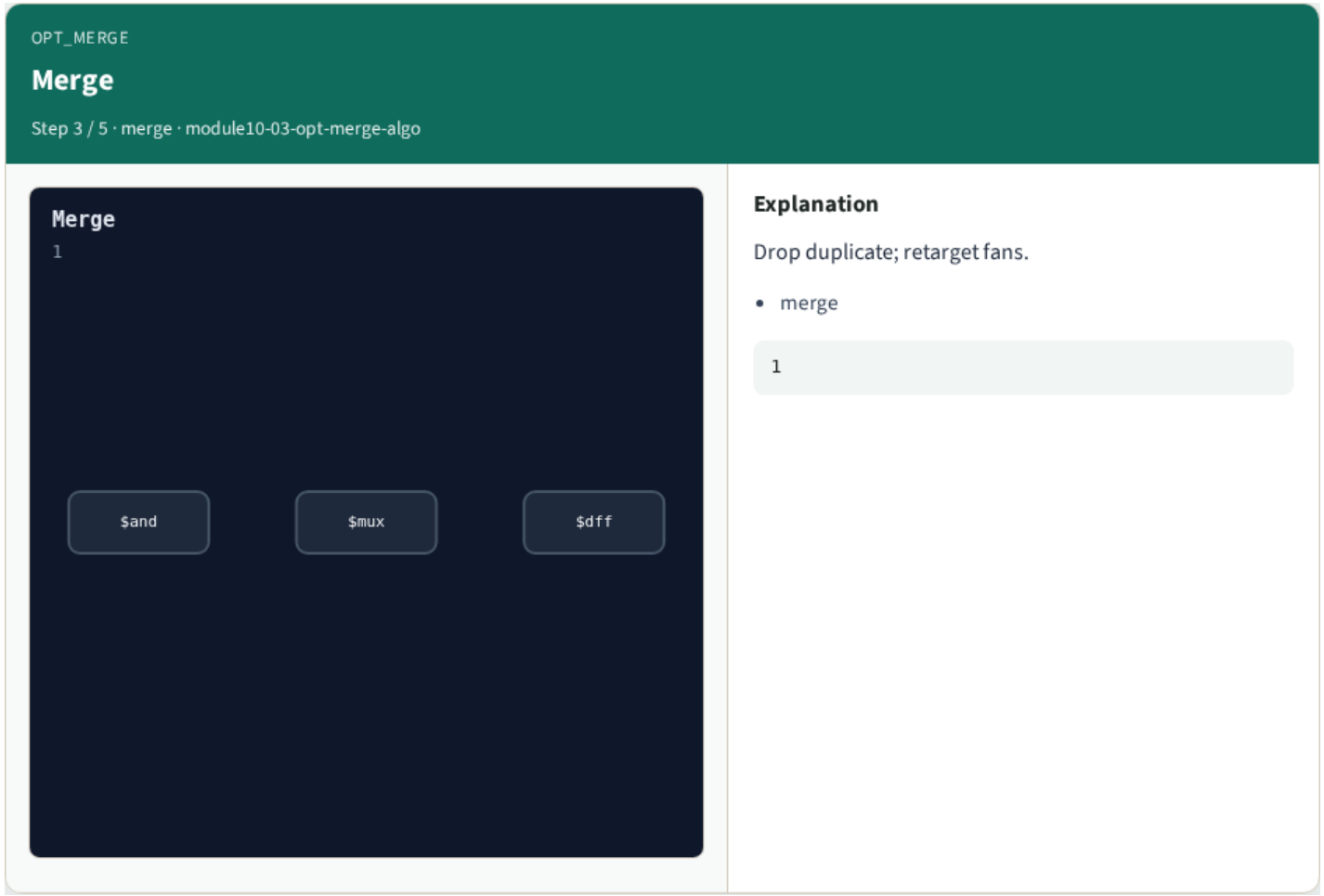
Duplicate AND

Hash key



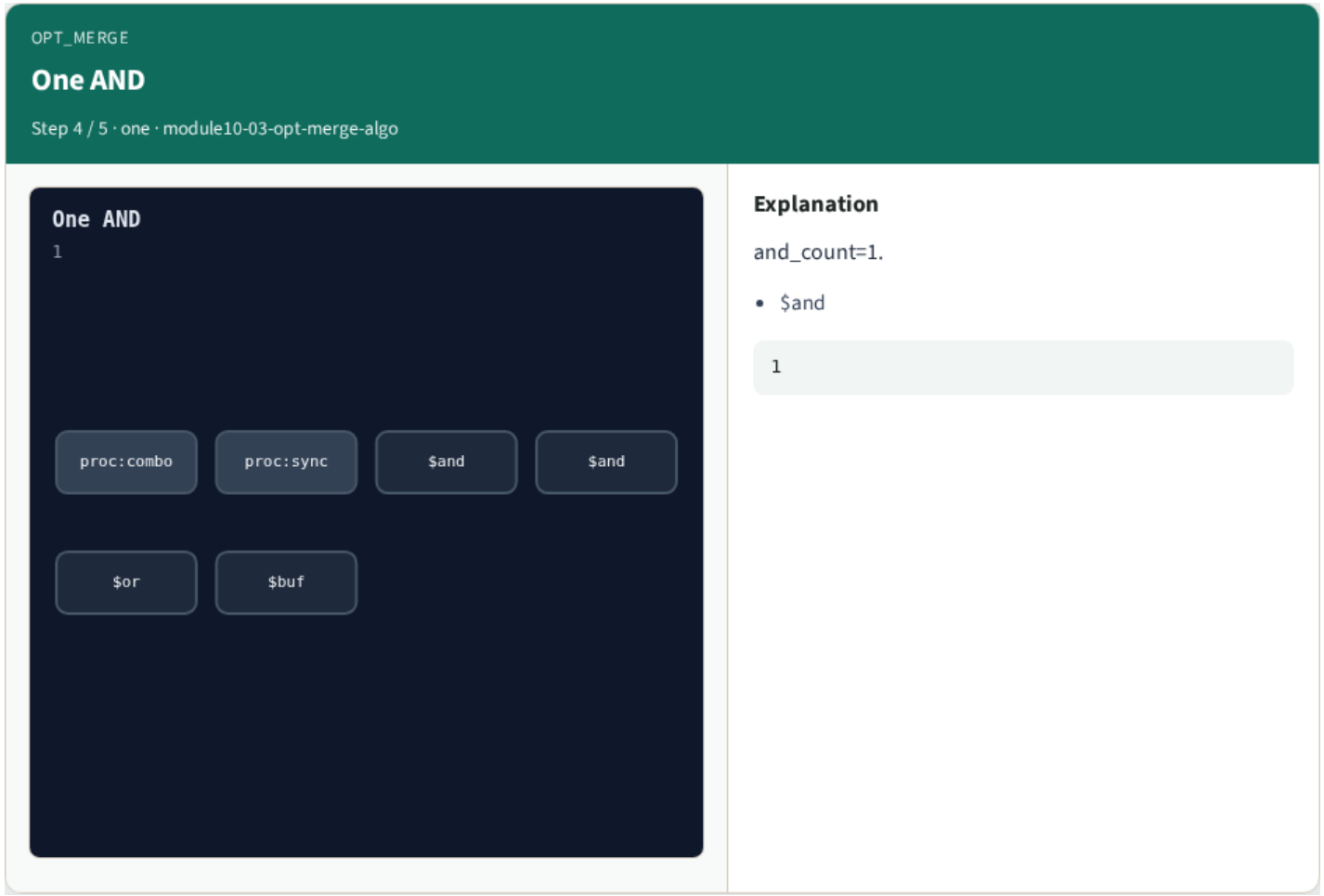
Hash key

Merge


OPT_MERGE
Merge
Step 3 / 5 · merge · module10-03-opt-merge-algo
Merge
1
$and
$mux
$dff
Explanation
Drop duplicate; retarget fans.
merge
1

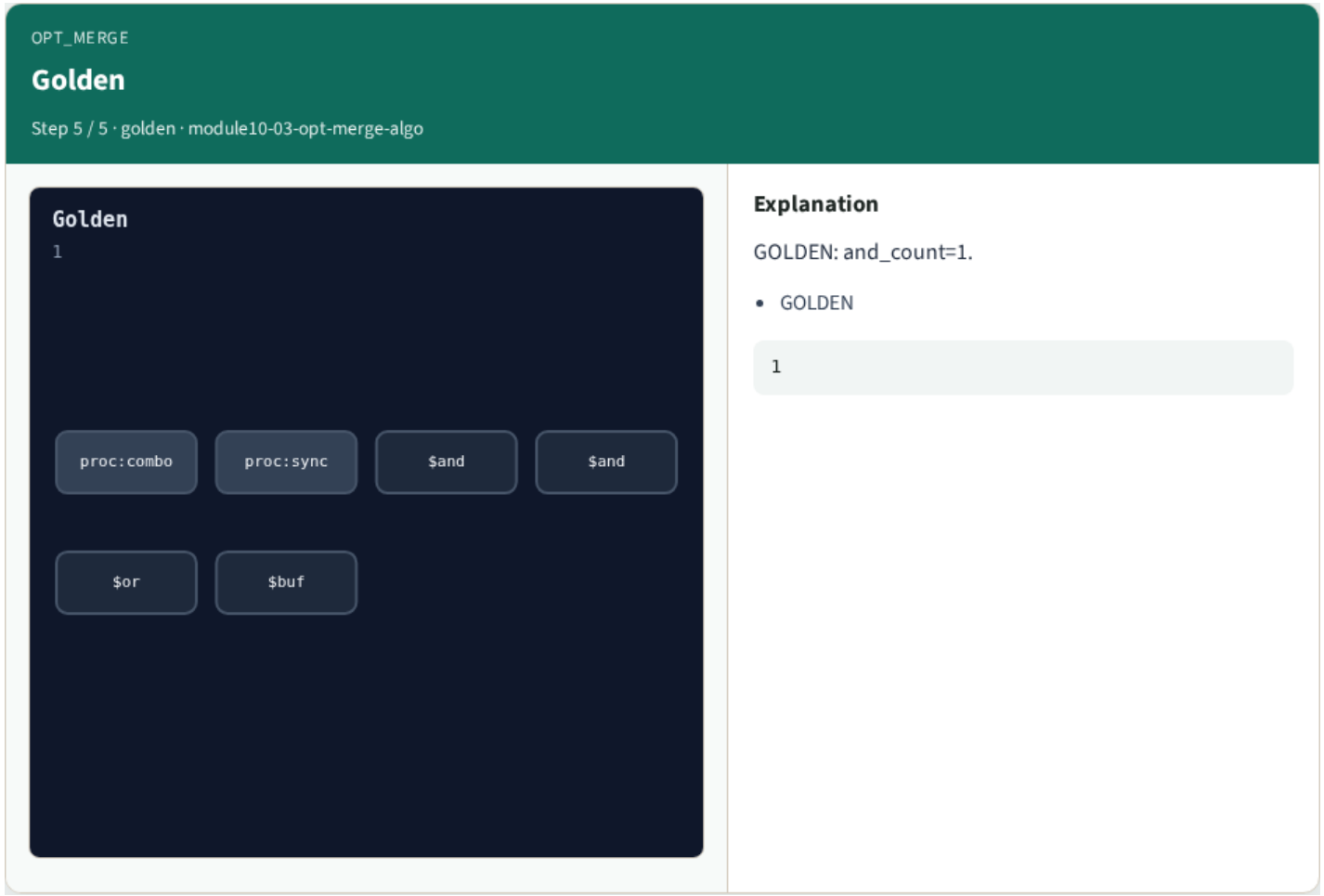
Merge

One AND



One AND

Golden



Golden

Browser lab track

- Open **opt-merge-algo**
- Merge the twin ANDs and Check and_count

Implement track

- Call algo_opt_merge(); assert and_count == 1

Pitfalls

- Merging cells with different inputs
- Forgetting to remap fanins

Try these

- From the common folder, run the algo solvers self-test, or call `algo_opt_merge` directly

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
after opt_clean
key ← (type,A,B) for and/or
if key seen: remap Y and drop duplicate
ok if and_count == 1
```

Next

- Complete the quiz for this part
- Next: techmap with a map file